

Carbon Farm

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Figure 1: Carbon Farm.

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1 Abstract

Contribute your CO₂ to our Carbon Garden, where it is transformed into nourishment for algae cultures, which are then processed into bioplastics uniquely made from your breath.

To participate, simply speak or breathe into our CO₂ collector. Choose from prompts like the Paris Climate Accord, corporate greenwashing claims, or a personal confession of your carbon footprint. You'll receive real-time feedback showing your CO₂

emissions and the corresponding algae growth. Our system contextualizes your contribution: how many breaths did it take for your commute? How many breaths are in the burger you ate? How many breaths power a plane? We turn the hot-air statements of governments and corporations into a grassroots ingredient for bio-production.

As you breathe into the Carbon Garden, the algae reward you with a live bubble orchestra—the sound of carbon sequestering and oxygen releasing. The harvested algae are then converted into eco-friendly bioplastics. As a token, you receive a bioplastic badge with your last name. This exhibit is part of a larger exploration of our connection to CO₂, aiming to give people a tangible sense of their carbon impact.

Beyond its artistic value, the Carbon Garden also functions as a household-scale carbon sequestration unit, akin to a rain catcher. Integrating Carbon Gardens into homes can capture atmospheric CO₂ and convert it into filament for 3D printing. This is our vision for the future: sustainable bioproduction fostering a new era of carbon-negative creation at home.

While carbon removal (CDR) technologies like this are vital for future sustainability, they must be paired with significant emissions mitigation efforts. For perspective, capturing your breath into the Carbon Garden for a year would only offset the carbon footprint of one cross-country flight or 27,500 breaths for a single burger.

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